

STOR 155.010 Lecture 08

Linear Regression II

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Overview

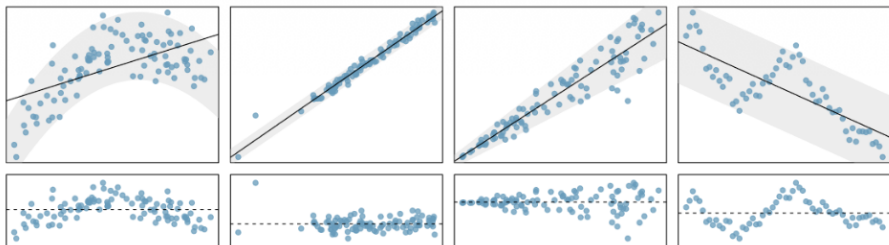
- 1 Conditions for fitting a linear model
- 2 Outliers
- 3 Extrapolation

Conditions for fitting a linear model

When fitting a least squares line, we generally require:

- **Linearity:** The data should show a linear trend.
- **Nearly normal residuals:** Generally, the residuals must be nearly normal. When this condition is found to be unreasonable, it is usually because of outliers or concerns about influential points.
- **Constant variability:** The variability of points around the least squares line remains roughly constant.
- **Independent observations:** Time series data (such as stock price each day) may have an underlying structure that should be considered in a model and analysis.

Different Scenarios



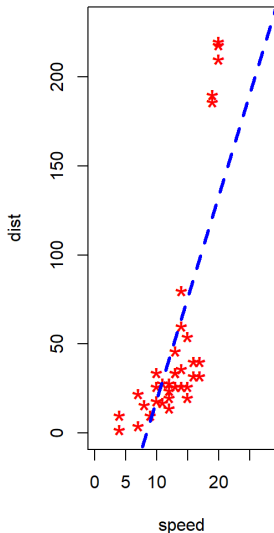
- 1 Linearity fails
- 2 Outliers exist
- 3 Errors not independent of x
- 4 Dependent observations

Overview

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Example

With Outliers



Outliers in Regression

Residual outlier

- Far from the line in y -direction (Typical x , unusual y).

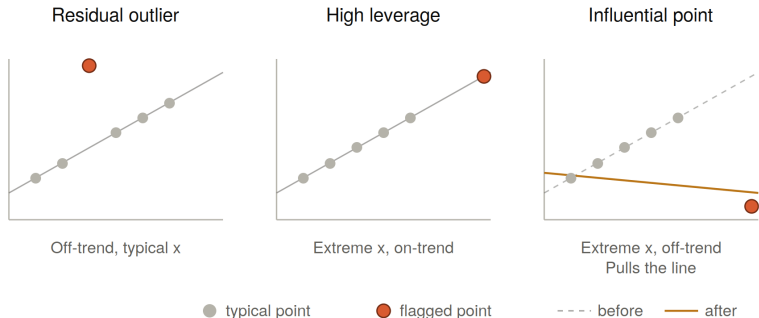
High leverage point

- Far from others in x -direction (Extreme x_i , far from $\bar{x}$).
- Potential to pull the line, but not guaranteed to.

Influential point

- High leverage **and** noticeably shifts the fitted line.
- Removing it changes slope/intercept meaningfully.

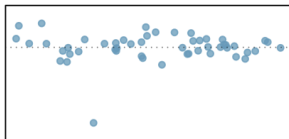
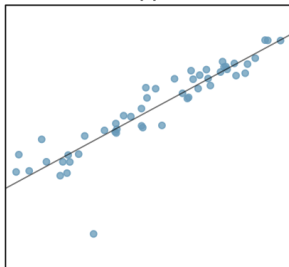
More on Outliers



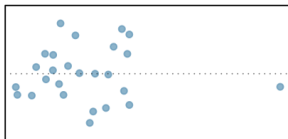
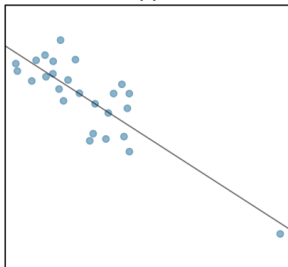
- Plot first: leverage and residual size together determine influence, not either alone.
- Check if it's a data-entry error, a rare-but-real case, or genuine noise.
- Report results with and without the point if removing it changes your conclusion; never delete silently.

Practice

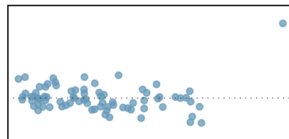
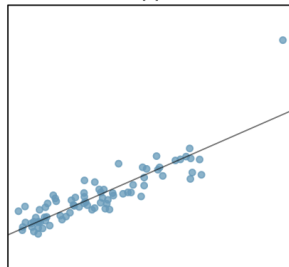
(1)



(2)

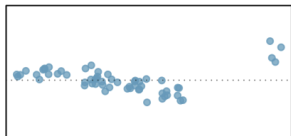
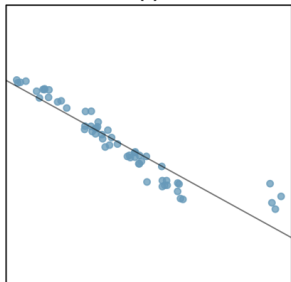


(3)

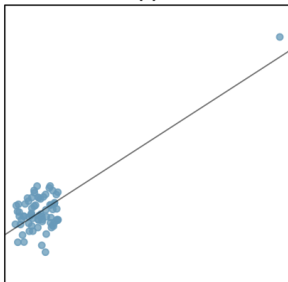


Practice (contd.)

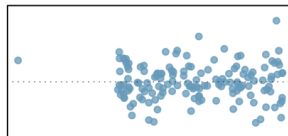
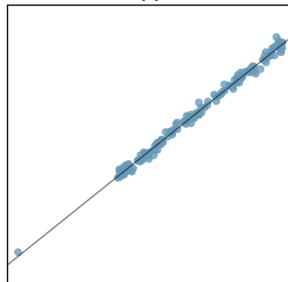
(4)



(5)



(6)



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Example

“When those blizzards hit the East Coast this winter, it proved to my satisfaction that global warming was a fraud. That snow was freezing cold. But in an alarming trend, temperatures this spring have risen.

Consider this: On February 6th it was 10 degrees. Today it hit almost 80. At this rate, by August it will be 220 degrees. So clearly folks the climate debate rages on.”

-Stephen Colbert (April 6th, 2010)

Definition

